

Path-spread-resolved quantum transition-state theory for proton transfer

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Centroid quantum transition-state theory and ring-polymer molecular dynamics yield thermal rates, but they collapse the imaginary-time path of the transferring proton onto a single collective coordinate. We define the proton-path spread χ_p as the mean-square bead displacement of the ring polymer along the proton-sharing coordinate—the squared radius of gyration projected onto that coordinate—and use it to resolve the centroid dividing-surface ensemble by imaginary-time geometry. The associated path-spread-resolved (QRR) formula is a decomposition of the usual centroid flux, not a more accurate dynamical rate. For a harmonic proton mode, $\langle\chi_p\rangle$ has a closed continuum formula and an exact finite-bead counterpart; Levy-staging path-integral Monte Carlo reproduces the finite-bead result to within 1%. The H/D ratio of $\langle\chi_p\rangle$ interpolates between $\sqrt{m_D/m_H}$ as $T \rightarrow 0$ and m_D/m_H as $T \rightarrow \infty$. Exact Eckart rates at 300 K give a tunneling correction $\kappa_H = 1.28$ –2.68 and a kinetic isotope effect of 1.68–2.28 as the barrier rises from 4 to 16 $k_B T$, against a classical isotope factor of $\sqrt{2}$. On an asymmetric double well, umbrella sampling shows that the dividing-surface path-spread distribution for H is systematically broader than for D ($f_{\text{deloc}}^H \approx 0.18$ –0.23 versus $f_{\text{deloc}}^D \approx 0.01$ –0.04). We do not report a QRR rate, nor a condensed-phase application. The estimator is a collective variable that can be accumulated in any existing path-integral code.

Keywords: proton transfer, path-integral molecular dynamics, quantum transition-state theory, kinetic isotope effect, ring-polymer molecular dynamics

I. INTRODUCTION

Proton transfer, and proton-coupled electron transfer (PCET) in particular, is a setting in which nuclear quantum effects, solvent reorganization, and, at electrodes, the applied potential jointly control the rate.^{1–4} Kinetic isotope effects (KIEs) and non-Arrhenius temperature dependence are the usual experimental signatures of tunneling,^{4,5} but they do not by themselves identify the imaginary-time path geometry that carries the reactive flux.

Path-integral methods address that geometry at the level of the thermal density matrix.^{6–8} Centroid quantum transition-state theory (QTST) replaces the classical reaction coordinate by the ring-polymer centroid and evaluates a flux through a dividing surface in centroid space.^{9–12} Ring-polymer molecular dynamics (RPMD) propagates the same polymer in real time and supplies a recrossing correction to the centroid flux.^{13–15} Instanton theory, in the complementary deep-tunneling limit, optimizes a single dominant imaginary-time orbit.^{16,17} Transition-path sampling retains full real-time path information, at a cost that is not organized by equilibrium imaginary-time geometry.¹⁸

What these approaches do not provide is a cheap, equilibrium order parameter that distinguishes compact ring polymers (thermally activated, localized near the donor

or acceptor) from delocalized polymers whose beads span the transfer coordinate (tunneling-like imaginary-time geometry). The ring-polymer radius of gyration is already used as a diagnostic of delocalization in condensed-phase proton studies.^{19,20} Here we restrict that diagnostic to the chemical sharing coordinate, give it exact harmonic limits, and use it to *condition* the QTST flux.

The resulting path-spread-resolved rate, which we call QRR, is

$$k_{\text{QRR}} = \int_0^\infty d\chi_p k_{\text{QTST}}(\chi_p) \kappa(\chi_p) p^\ddagger(\chi_p), \quad (1)$$

where $p^\ddagger(\chi_p)$ is the distribution of path spreads at the dividing surface and $\kappa(\chi_p)$ is a class-conditional transmission coefficient. When κ is taken independent of χ_p , Eq. (1) reduces identically to ordinary QTST. QRR is therefore a decomposition of a known estimator, not a claim of a more accurate thermal rate. Its value is mechanistic: it reports which imaginary-time geometries carry the flux, how that mix depends on isotope and barrier, and whether path spread couples to a solvent coordinate.

Section II defines χ_p and the conditional free energy, states the solvable limits, and writes the flux decomposition. Section III describes the samplers. Section IV reports harmonic, Eckart, and double-well calculations. Section V discusses scope and limitations. We do not report condensed-phase electrochemical simulations; those require a validated reactive potential and are left as an application of the estimator.

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II. THEORY

A. Path-spread functional

The P -bead ring-polymer representation of the canonical density is standard.^{8,10} For a proton transferring between donor and acceptor oxygens we use the bead-resolved sharing coordinate

$$\xi_p^{(s)} = r_{\text{ODH}}^{(s)} - r_{\text{OAH}}^{(s)}, \quad \bar{\xi}_p = P^{-1} \sum_{s=1}^P \xi_p^{(s)}. \quad (2)$$

The path-spread functional is the bead variance along that coordinate,

$$\chi_p = \frac{1}{P} \sum_{s=1}^P (\xi_p^{(s)} - \bar{\xi}_p)^2. \quad (3)$$

This is the squared radius of gyration of the polymer projected onto ξ_p . It is identically zero at $P = 1$ and is non-negative for all configurations. Compact paths have χ_p of order the zero-point spread in a well; delocalized paths have χ_p large enough that beads occupy both sides of the barrier.

The joint density $P(\bar{\xi}_p, \chi_p)$ defines the conditional free energy

$$F(\bar{\xi}_p, \chi_p) = -k_B T \ln P(\bar{\xi}_p, \chi_p) + C. \quad (4)$$

Projection onto $\bar{\xi}_p$ recovers the centroid potential of mean force.¹⁰ Slices at fixed $\bar{\xi}_p$ resolve path class, as sketched in Fig. 1. A solvent or energy-gap coordinate λ_s may be added without changing the construction; the three-dimensional surface $F(\bar{\xi}_p, \chi_p, \lambda_s)$ is the natural object for PCET, but the calculations below are two-dimensional unless stated.

B. Flux decomposition

Let $\bar{\xi}_p^\ddagger$ locate the dividing surface. The marginal

$$p^\ddagger(\chi_p) = \frac{e^{-\beta F(\bar{\xi}_p^\ddagger, \chi_p)}}{\int d\chi_p' e^{-\beta F(\bar{\xi}_p^\ddagger, \chi_p')}} \quad (5)$$

is the probability that a dividing-surface configuration belongs to path class χ_p . The class-conditional QTST flux is the usual positive-velocity centroid flux restricted to that class,

$$k_{\text{QTST}}(\chi_p) = Z_R^{-1} \langle [\dot{\xi}]_+ \delta(\bar{\xi}_p - \bar{\xi}_p^\ddagger) \delta(\chi_p[\mathbf{R}] - \chi_p) \rangle. \quad (6)$$

A Bennett–Chandler transmission coefficient $\kappa(\chi_p)$ may be attached to each class.^{14,21,22} Integrating over χ_p produces Eq. (1). We emphasize two restrictions. First, $\kappa(\chi_p)$ computed from a Wigner series in $\hbar^2$ is a tunneling correction to a parabolic barrier,^{23,24} not a recrossing

factor; the two must not be conflated. Second, for highly delocalized paths the Wigner series is not a controlled approximation, and a class-conditional RPMD flux-side correlation is the consistent dynamical correction.¹⁴ The numerical work below reports $p^\ddagger(\chi_p)$ and related fractions, which are equilibrium properties, and uses exact Eckart rates when a thermal rate is required.

C. Solvable limits

Remark 1 (Classical limit) At $P = 1$, Eq. (3) gives $\chi_p = 0$ identically, so $p^\ddagger(\chi_p) = \delta(\chi_p)$ and k_{QRR} reduces to classical transition-state theory. This is a consistency check, not an independent result.

Theorem 1 (Harmonic continuum) For $V = \frac{1}{2}m\omega^2 q^2$,

$$\langle \chi_p \rangle_\infty = \frac{\hbar}{2m\omega} \coth\left(\frac{\beta\hbar\omega}{2}\right) - \frac{k_B T}{m\omega^2}. \quad (7)$$

The first term is the equal-time imaginary-time correlator $G(0)$; the second is the centroid variance, which is classical for a harmonic oscillator.¹⁰ The difference is non-negative because $\coth x \geq 1/x$ for $x > 0$. As $T \rightarrow 0$, $\langle \chi_p \rangle \rightarrow \hbar/(2m\omega)$. As $T \rightarrow \infty$, the leading term is the Wigner correction $\hbar^2\beta/(12m)$, independent of ω .²³ The Matsubara-mode derivation is given in the supplementary material.

Proposition 1 (Exact finite P) For the same oscillator discretized with P beads,

$$\langle \chi_p \rangle_P = \frac{1}{P} \sum_{k=1}^{P-1} \frac{1}{\beta\lambda_k}, \quad \lambda_k = \frac{4mP}{\hbar^2\beta^2} \sin^2\left(\frac{\pi k}{P}\right) + \frac{m\omega^2}{P}. \quad (8)$$

As $P \rightarrow \infty$, $\langle \chi_p \rangle_P \rightarrow \langle \chi_p \rangle_\infty$.

Equation (8) is the quantity that a primitive path-integral estimator of Eq. (3) actually samples. Comparing Monte Carlo to Eq. (7) at modest P therefore mixes statistical error with Trotter error. We compare to Eq. (8) first.

Corollary 1 (Harmonic isotope scaling) On a shared harmonic surface ($k = m\omega^2$ fixed), $\langle \chi_p \rangle_H > \langle \chi_p \rangle_D$ for all $T > 0$. The ratio $\mathcal{R} = \langle \chi_p \rangle_H / \langle \chi_p \rangle_D$ satisfies

$$\mathcal{R}(T \rightarrow 0) = \sqrt{m_D/m_H}, \quad \mathcal{R}(T \rightarrow \infty) = m_D/m_H. \quad (9)$$

The low-temperature limit is the ratio of zero-point amplitudes. The high-temperature limit is the ratio of leading Wigner corrections. Both follow from Eq. (7) and do not extend automatically to anharmonic barriers.

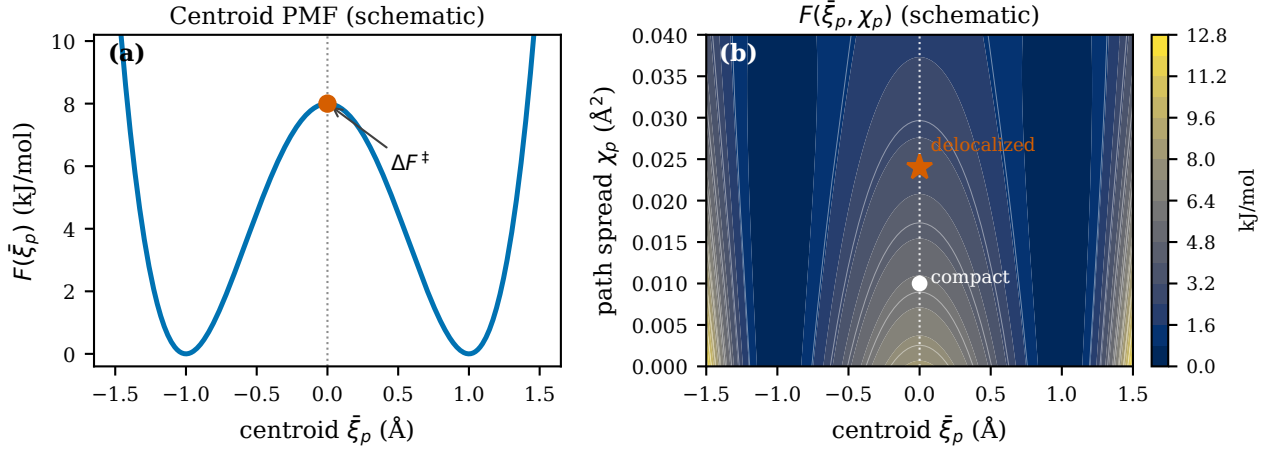


FIG. 1. Schematic, not computed data. (a) Centroid potential of mean force $F(\bar{\xi}_p)$, in which all ring-polymer shapes at a given centroid are averaged. (b) Conditional free energy $F(\bar{\xi}_p, \chi_p)$. Compact and delocalized dividing-surface geometries that share $\bar{\xi}_p = 0$ are marked.

Proposition 2 (Factorization) *If $F(\bar{\xi}_p, \chi_p, \lambda_s) = F_p(\bar{\xi}_p, \chi_p) + F_s(\lambda_s)$, then $p^\ddagger(\chi_p)$ is independent of λ_s and the χ_p integral in Eq. (1) reduces to ordinary QTST when κ is class-independent.*

A nonzero mixed derivative $\partial^2 F / \partial \chi_p \partial \lambda_s$, or a nonzero dividing-surface correlation $\text{corr}(\chi_p, \lambda_s)$, would then signal proton-solvent coupling beyond a product Marcus picture.^{2,3} We do not claim to have measured that correlation with statistical control in this work.

III. METHODS

Path-integral Monte Carlo uses Levy staging of the free-particle bridge,^{25,26} which samples the ring-polymer springs exactly, together with a Metropolis decision on the physical potential and a rigid centroid move. The primitive estimator of χ_p is the bead variance. For the harmonic oscillator we also evaluate Eqs. (7) and (8) in closed form.

Asymmetric double-well (ADW) free energies are obtained from one-dimensional umbrella sampling along $\bar{\xi}_p$ ²⁷ with harmonic restraints, reweighted by the weighted histogram analysis method (WHAM).²⁸ Two-dimensional histograms of $(\bar{\xi}_p, \chi_p)$ use the same WHAM weights. The delocalized fraction at the dividing surface is

$$f_{\text{deloc}} = \int_{\chi_p \geq 2\chi_c} p^\ddagger(\chi_p) d\chi_p, \quad (10)$$

with the threshold $\chi_c = \hbar / (2m\omega^*)$ evaluated at $\omega^* = 1000 \text{ cm}^{-1}$, giving $\chi_c \approx 0.010 \text{ \AA}^2$. Compact and intermediate classes use $0.5\chi_c$ and $2\chi_c$ as bookkeeping cuts; the continuous $p^\ddagger(\chi_p)$ is the primary object.

Exact thermal rates for the symmetric Eckart barrier $V(x) = V_0 \text{sech}^2(x/a)$ are

$$k = \frac{1}{2\pi\hbar Q_r} \int_0^\infty dE e^{-\beta E} T(E), \quad (11)$$

with $T(E)$ the exact transmission probability of the symmetric Eckart barrier²⁹ and $Q_r = (m/2\pi\hbar^2\beta)^{1/2}$. Classical TST is $k_{\text{TST}} = \sqrt{k_B T / (2\pi m)} e^{-\beta V_0}$. The Wigner estimate multiplies k_{TST} by $1 + (\beta\hbar\omega_b)^2/24$, where $\omega_b = \sqrt{2V_0/(ma^2)}$. All Eckart integrals use a logarithmically stable evaluation of $T(E)$.

Temperatures are 300 K unless noted. Masses are 1 and 2 amu. Random number generators are seeded at 42. Code and JSON outputs are in the repository accompanying this article.

IV. RESULTS

A. Harmonic oscillator

Figure 2 compares Eq. (7) with Levy-staging Monte Carlo for a 1000 cm^{-1} mode over 100–700 K. At 300 K the continuum values are $\langle \chi_p \rangle_H = 0.01011 \text{ \AA}^2$ and $\langle \chi_p \rangle_D = 0.00572 \text{ \AA}^2$. With $P = 64$ (H) and $P = 32$ (D), the exact finite-bead formula differs from the continuum by 0.12% and 0.29%, respectively, and the Monte Carlo root-mean-square relative error against Eq. (8) is 0.56% (H) and 1.03% (D).

The isotope ratio $\mathcal{R}(T)$ increases from 1.52 at 100 K toward $\sqrt{2}$ as T is lowered, and from 1.77 at 300 K to 1.94 at 700 K, approaching 2 from below, as required by Corollary 1 [Fig. 2(b)]. Figure 2(c) shows that the primitive estimator tracks Eq. (8) from $P = 4$ to $P = 128$ and that the Trotter error relative to the continuum is already below 0.5% at $P = 32$ for this frequency and temperature.

A staging move that draws Brownian-bridge *marginals* independently, rather than the sequential Lévy construction, produces a spurious increase of $\langle\chi_p\rangle$ with P and should not be used.

B. Eckart barrier

The symmetric Eckart barrier with width $a = 0.50 \text{ \AA}$ provides thermal rates that do not depend on path-integral sampling. Figure 3(a) shows the exact tunneling correction $\kappa = k/k_{\text{TST}}$ at 300 K. For H, κ increases from 1.28 at $V_0 = 4 k_{\text{B}}T$ to 2.68 at $V_0 = 16 k_{\text{B}}T$; the corresponding Wigner factors are 1.22 and 1.86. Wigner is therefore a useful first correction at moderate $\hbar\omega_b/k_{\text{B}}T$ and underestimates the exact κ once $\hbar\omega_b/k_{\text{B}}T \gtrsim 4$. Deuterium remains closer to TST, as expected from the lower barrier frequency.

The exact KIE [Fig. 3(b) and Table I] is 1.68 at $4 k_{\text{B}}T$ and 2.28 at $16 k_{\text{B}}T$. Classical TST contributes only $\sqrt{m_D/m_H} = \sqrt{2} \approx 1.414$, independent of barrier height. The Wigner KIE tracks the exact result qualitatively (1.55 to 1.84) but lags at the highest barriers. These numbers set the scale of the nuclear quantum effect that a path-spread diagnostic is meant to resolve. They are *not* QRR rates: the Eckart calculation is exact one-dimensional scattering, used here as an independent benchmark of when tunneling matters.

TABLE I. Symmetric Eckart barrier at $T = 300 \text{ K}$, $a = 0.50 \text{ \AA}$. $\kappa = k/k_{\text{TST}}$. Classical KIE is $\sqrt{2}$ at every barrier.

$V_0/k_{\text{B}}T$	$\hbar\omega_b^H/k_{\text{B}}T$	κ_H	κ_D	KIE (exact)	KIE (Wigner)
4	2.27	1.28	1.08	1.68	1.55
8	3.22	1.60	1.28	1.77	1.67
12	3.94	2.04	1.36	2.12	1.76
16	4.55	2.68	1.66	2.28	1.84

C. Asymmetric double well

The ADW $V(q) = V_0[(q/q_0)^2 - 1]^2 + \alpha q$ with $q_0 = 0.70 \text{ \AA}$ is a bound proton-transfer model. Figure 4(a) shows the WHAM centroid PMF at $V_0 = 8 k_{\text{B}}T$: the H barrier is 15.9 kJ mol^{-1} and the D barrier is 17.6 kJ mol^{-1} . Figure 4(b) is the quantity QRR actually uses, the dividing-surface density $p^\ddagger(\chi_p)$. It is broader for H than for D, with means $\langle\chi_p\rangle_\ddagger^H = 0.015 \text{ \AA}^2$ and $\langle\chi_p\rangle_\ddagger^D = 0.008 \text{ \AA}^2$. With the bookkeeping cut $2\chi_c = 0.020 \text{ \AA}^2$, the delocalized weight is $f_{\text{deloc}}^H = 0.18\text{--}0.23$ across $V_0 = 2\text{--}8 k_{\text{B}}T$, versus $f_{\text{deloc}}^D = 0.014\text{--}0.045$ [Fig. 4(c) and Table II]. For H the fraction is essentially flat in this barrier window; the robust observation is the isotope contrast, not a strong barrier trend.

These are equilibrium path-class statistics, not rates. We do not convert f_{deloc} into a QRR rate or into a

claimed improvement over RPMD. Two-dimensional histograms of $F(\xi_p, \chi_p)$ from the same umbrellas are too sparsely sampled in χ_p to publish as free-energy surfaces.

TABLE II. Asymmetric double well at $T = 300 \text{ K}$. Centroid barriers from 1D WHAM; f_{deloc} from the WHAM-weighted dividing-surface ensemble with $2\chi_c = 0.020 \text{ \AA}^2$.

$V_0/k_{\text{B}}T$	ΔF_H (kJ/mol)	ΔF_D (kJ/mol)	f_{deloc}^H	f_{deloc}^D	$\langle\chi_p\rangle_\ddagger^H$ ($\text{\AA}^2$)	$\langle\chi_p\rangle_\ddagger^D$ ($\text{\AA}^2$)
2	4.25	4.60	0.18	0.014	0.014	0.007
4	8.90	9.02	0.23	0.020	0.015	0.007
6	12.4	14.3	0.22	0.024	0.015	0.007
8	15.9	17.6	0.23	0.045	0.015	0.008

D. What is not shown

A bilinear proton-solvent model $V = V_{\text{ADW}}(q_p) + \frac{1}{2}k_s q_s^2 + \kappa_c q_p q_s$ was sampled without umbrellas. Dividing-surface correlations $\text{corr}(\chi_p, q_s)$ lie between -0.06 and -0.16 with a few hundred crossings and do not trend cleanly with κ_c . That scan is not evidence for or against Proposition 2; a controlled test would require umbrella sampling in $(\xi_p, \chi_p, \lambda_s)$. We therefore omit it from the figures.

V. DISCUSSION

QRR does not replace RPMD, instanton theory, or vibronic PCET rate laws.^{2,14,17,30} RPMD remains the appropriate dynamical correction to centroid QTST in the crossover regime; instanton theory remains the appropriate semiclassical description when a single tunneling orbit dominates; Hammes-Schiffer vibronic theory remains the appropriate nonadiabatic framework when electron-proton nonadiabaticity is essential. What χ_p adds is an equilibrium resolution of imaginary-time geometries that those methods otherwise average or optimize.

The present calculations do not evaluate Eq. (1) as a rate. Doing so would require a class-conditional transmission coefficient, which we have not computed. The Eckart section reports exact scattering rates; the ADW section reports $p^\ddagger(\chi_p)$. Connecting the two is an analogy, not a numerical identity.

Two condensed-phase hypotheses follow, and should be treated as such. First, if an increase of the observed KIE with driving force^{30,31} is carried by tunneling-like paths, $\Delta f_{\text{deloc}} = f_{\text{deloc}}^H - f_{\text{deloc}}^D$ should increase with that driving force. Second, a breakdown of $F = F_p + F_s$ at the dividing surface would be a signature of proton-solvent coupling beyond a product Marcus picture. Neither hypothesis is tested here.

Limitations are sharp. χ_p is an imaginary-time polymer statistic, not a spectroscopic observable. The cut

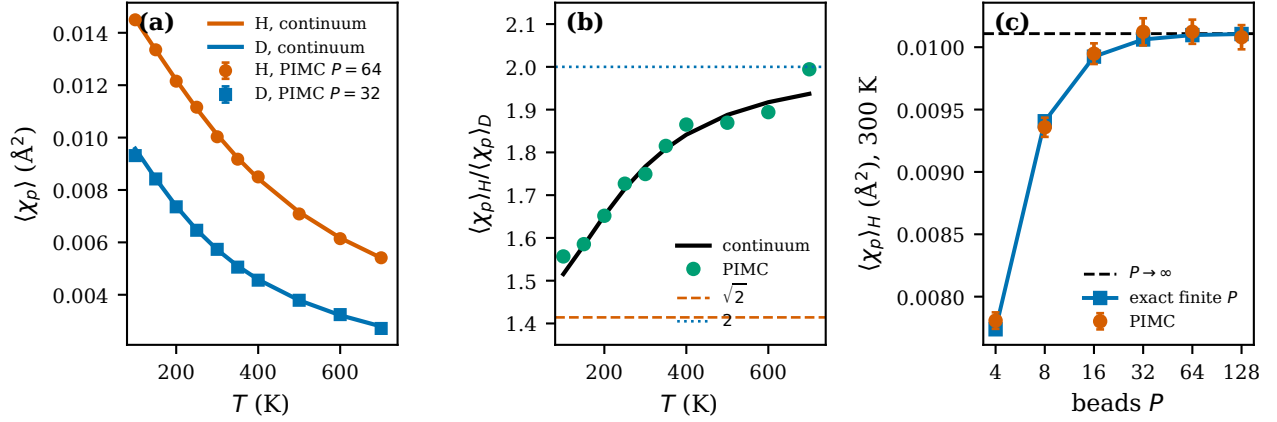


FIG. 2. Harmonic path spread for $\omega = 1000 \text{ cm}^{-1}$. (a) $\langle\chi_p\rangle(T)$ from Eq. (7) (lines) and Levy-staging PIMC (symbols). (b) Isotope ratio, with the $T \rightarrow 0$ limit $\sqrt{2}$ and the $T \rightarrow \infty$ limit 2. (c) Bead convergence of $\langle\chi_p\rangle_H$ at 300 K: PIMC versus the exact finite- P formula, Eq. (8).

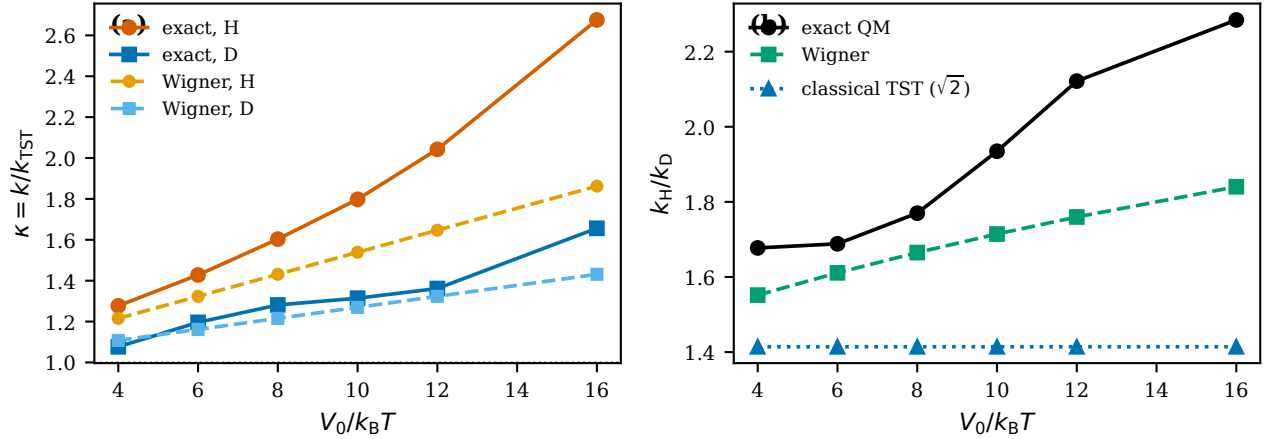


FIG. 3. Symmetric Eckart barrier, $a = 0.50 \text{ Å}$, $T = 300 \text{ K}$. (a) Exact and Wigner tunneling corrections versus barrier height. (b) Kinetic isotope effect. Classical TST is $\sqrt{2}$ at every V_0 .

χ_c is bookkeeping; Fig. 4(b) is the primary object. Recrossing is not resolved by class. The models are one-dimensional. Finite- P errors in χ_p are under control for the 1000 cm^{-1} mode at $P \geq 32$; stiffer stretches will need more beads or a virial estimator.

VI. CONCLUSIONS

We have defined a projected ring-polymer path spread χ_p for proton transfer, derived its harmonic continuum and finite-bead limits, and used it to resolve the centroid dividing-surface ensemble. Levy-staging Monte Carlo reproduces the exact finite-bead harmonic formula to about one percent. Exact Eckart rates quantify when tunneling corrections and KIEs depart from TST. On an asymmetric double well, the dividing-surface path-spread distribution is robustly broader for H than for D. Equation (1) is

a bookkeeping decomposition of QTST; it is not shown here to be a better rate. The estimator is a collective variable and can be accumulated in any path-integral molecular dynamics run.

ACKNOWLEDGMENTS

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.22718950> (Ref. 32). The source repository is <https://github.com/vmercel/QRR-PCET>.

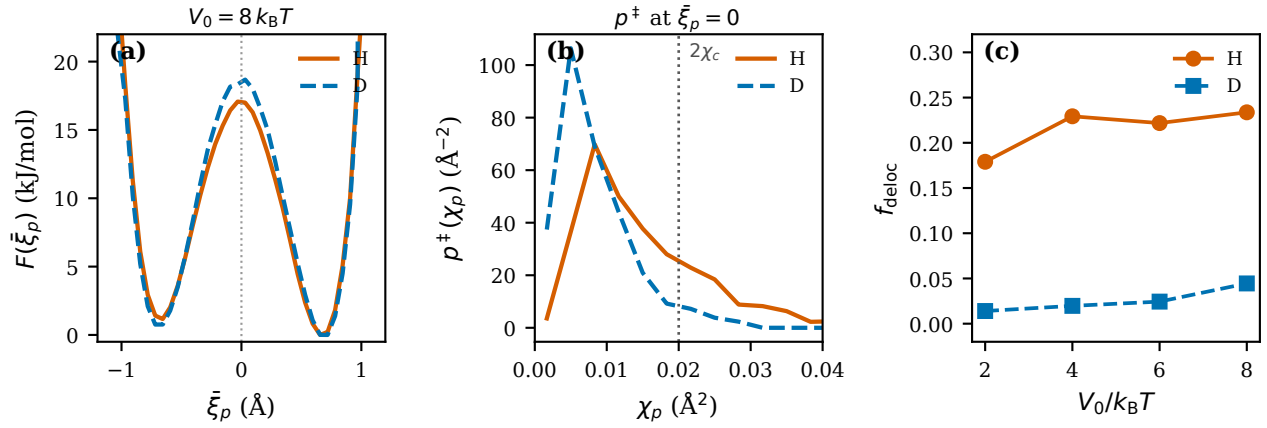


FIG. 4. Asymmetric double well, $T = 300$ K. (a) Centroid PMF at $V_0 = 8 k_B T$. (b) Dividing-surface path-spread density $p^+(\chi_p)$ at the same barrier; the dotted line is the bookkeeping cut $2\chi_c$. (c) Delocalized fraction versus barrier height. The H fraction is nearly flat; the D fraction remains an order of magnitude smaller.

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